

## ALGAZY OSTROV, Kazakhstan

WMO: 366860

Lat: 46.543N

Lon: 76.862E

Elev: 1145

StdP: 14.10

Time Zone: 6.00 (E06)

Period: 02-14

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.1      | -12.2 | -21.9                         | 1.7 | -15.3 | -18.1 | 2.1 | -11.9 | 31.9                  | 6.3  | 26.7 | 10.1 | 3.7                   | 50   | 0.594 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.8                   | 91.7            | 66.8 | 88.6 | 66.7 | 85.9 | 66.4 | 72.1                | 83.7 | 70.2 | 81.7 | 68.9 | 80.8 | 10.4                 | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.2                            | 108.0 | 79.4 | 66.2 | 100.7 | 78.0 | 64.4 | 94.4 | 76.9 | 36.7          | 83.7 | 35.0 | 82.0 | 33.9 | 80.7 | 80.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               | Max  | Min  | Max                        | Min   | Max  | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 27.0              | 22.8 | 20.7 | DB                         | -20.1 | 98.5 | 5.0                | 3.5                                                | -23.7     | 101.1 | -26.6      | 103.1 | -29.4      | 105.1 | -33.0      | 107.7 |
|                   |      |      | WB                         | -20.4 | 75.2 | 5.0                | 2.7                                                | -23.9     | 77.1  | -26.8      | 78.7  | -29.6      | 80.2  | -33.2      | 82.2  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 45.8   | 7.1   | 11.1  | 29.0  | 48.5  | 62.9  | 73.7  | 76.7  | 74.8  | 63.6  | 49.3  | 34.5  | 15.9  |
|                                                                                  | DBStd       | 25.92  | 10.41 | 11.72 | 11.07 | 8.59  | 6.78  | 5.83  | 5.09  | 5.56  | 7.25  | 6.87  | 8.15  | 11.59 |
|                                                                                  | HDD50       | 4818   | 1331  | 1090  | 650   | 126   | 2     | 0     | 0     | 0     | 2     | 96    | 464   | 1057  |
|                                                                                  | HDD65       | 8084   | 1796  | 1510  | 1115  | 497   | 121   | 7     | 1     | 3     | 112   | 486   | 914   | 1522  |
|                                                                                  | CDD50       | 3273   | 0     | 0     | 0     | 81    | 401   | 712   | 826   | 768   | 409   | 75    | 1     | 0     |
|                                                                                  | CDD65       | 1063   | 0     | 0     | 0     | 1     | 55    | 269   | 362   | 306   | 69    | 1     | 0     | 0     |
|                                                                                  | CDH74       | 8229   | 0     | 0     | 0     | 20    | 376   | 2091  | 3091  | 2311  | 338   | 2     | 0     | 0     |
|                                                                                  | CDH80       | 2673   | 0     | 0     | 0     | 2     | 91    | 679   | 1057  | 764   | 80    | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 8.8    | 6.6   | 6.5   | 7.9   | 8.5   | 9.7   | 9.4   | 9.6   | 9.6   | 10.0  | 9.3   | 9.8   | 7.8   |
| Precipitation                                                                    | PrecAvg     | 4.9    | 0.4   | 0.4   | 0.4   | 0.4   | 0.6   | 0.4   | 0.7   | 0.2   | 0.2   | 0.4   | 0.6   | 0.5   |
|                                                                                  | PrecMax     | 8.9    | 1.0   | 0.9   | 0.9   | 1.3   | 1.8   | 1.9   | 4.9   | 1.7   | 1.5   | 1.0   | 1.2   | 1.5   |
|                                                                                  | PrecMin     | 2.8    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   |
|                                                                                  | PrecStd     | 1.7    | 0.3   | 0.2   | 0.3   | 0.4   | 0.4   | 0.4   | 1.0   | 0.4   | 0.3   | 0.3   | 0.3   | 0.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.3  | 37.0  | 55.8  | 76.8  | 87.0  | 93.9  | 95.7  | 95.7  | 88.2  | 71.0  | 53.7  | 38.1  |
|                                                                                  |             | MCWB   | 32.8  | 33.9  | 44.9  | 56.0  | 63.0  | 65.7  | 68.3  | 67.8  | 62.8  | 54.1  | 48.9  | 35.7  |
|                                                                                  | 2%          | DB     | 30.2  | 33.9  | 49.6  | 69.6  | 82.3  | 89.9  | 91.6  | 90.6  | 81.4  | 65.5  | 50.0  | 34.5  |
|                                                                                  |             | MCWB   | 28.6  | 31.9  | 41.5  | 53.6  | 61.5  | 66.0  | 66.9  | 68.0  | 62.5  | 53.6  | 46.2  | 32.4  |
|                                                                                  | 5%          | DB     | 26.6  | 30.9  | 45.3  | 64.4  | 78.0  | 86.6  | 88.4  | 87.2  | 77.4  | 62.0  | 47.3  | 32.6  |
|                                                                                  |             | MCWB   | 25.4  | 28.9  | 39.2  | 51.8  | 59.8  | 66.2  | 66.8  | 67.5  | 60.7  | 52.8  | 43.7  | 30.7  |
|                                                                                  | 10%         | DB     | 22.1  | 27.8  | 41.8  | 60.6  | 74.3  | 83.7  | 85.6  | 84.2  | 74.3  | 59.1  | 44.7  | 30.7  |
|                                                                                  |             | MCWB   | 20.7  | 25.9  | 37.0  | 50.0  | 58.6  | 65.3  | 66.9  | 66.6  | 59.6  | 51.0  | 41.1  | 28.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 32.9  | 34.6  | 46.6  | 58.4  | 66.4  | 74.6  | 75.1  | 76.2  | 67.5  | 58.0  | 50.4  | 36.0  |
|                                                                                  |             | MCDB   | 33.9  | 36.6  | 53.6  | 71.1  | 80.3  | 86.4  | 87.7  | 88.3  | 80.6  | 64.8  | 52.8  | 37.1  |
|                                                                                  | 2%          | WB     | 28.9  | 31.8  | 42.7  | 55.5  | 63.9  | 70.5  | 72.0  | 71.9  | 64.8  | 55.5  | 46.8  | 32.9  |
|                                                                                  |             | MCDB   | 30.2  | 33.8  | 48.8  | 66.1  | 77.0  | 81.4  | 83.7  | 83.8  | 77.2  | 62.7  | 49.0  | 34.3  |
|                                                                                  | 5%          | WB     | 25.3  | 29.1  | 39.6  | 53.2  | 62.0  | 68.9  | 70.5  | 69.7  | 63.0  | 53.3  | 44.1  | 30.9  |
|                                                                                  |             | MCDB   | 26.6  | 31.0  | 44.4  | 62.5  | 73.9  | 80.7  | 82.0  | 81.3  | 74.7  | 60.1  | 47.1  | 32.4  |
|                                                                                  | 10%         | WB     | 20.8  | 26.0  | 37.3  | 51.0  | 60.1  | 67.4  | 69.2  | 68.1  | 61.0  | 51.6  | 41.5  | 29.0  |
|                                                                                  |             | MCDB   | 22.1  | 27.6  | 41.6  | 59.2  | 71.4  | 80.1  | 81.0  | 80.7  | 72.5  | 59.0  | 44.4  | 30.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.2  | 15.4  | 14.7  | 17.1  | 17.5  | 16.9  | 15.8  | 16.3  | 15.1  | 12.2  | 9.1   | 11.5  |
|                                                                                  |             | MCDBR  | 14.9  | 14.0  | 17.9  | 22.2  | 22.6  | 20.0  | 19.2  | 20.1  | 18.6  | 16.6  | 11.0  | 9.8   |
|                                                                                  |             | MCWBR  | 14.2  | 13.2  | 12.2  | 10.6  | 8.7   | 7.4   | 7.6   | 8.6   | 9.2   | 9.8   | 8.2   | 9.0   |
|                                                                                  | 5% WB       | MCDBR  | 14.3  | 13.9  | 17.0  | 19.3  | 20.0  | 17.3  | 16.6  | 17.9  | 17.0  | 14.6  | 10.6  | 9.6   |
|                                                                                  |             | MCWBR  | 13.7  | 13.3  | 12.2  | 10.3  | 9.5   | 8.4   | 8.6   | 9.6   | 9.9   | 10.3  | 8.4   | 9.2   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.320  | 0.353 | 0.401 | 0.443 | 0.447 | 0.447 | 0.453 | 0.407 | 0.371 | 0.373 | 0.351 | 0.318 |       |
|                                                                                  | taud        | 2.036  | 1.989 | 2.043 | 2.015 | 2.057 | 2.122 | 2.130 | 2.239 | 2.293 | 2.192 | 2.143 | 2.072 |       |
|                                                                                  | Ebn at Noon | 230    | 248   | 256   | 257   | 260   | 261   | 258   | 266   | 265   | 241   | 219   | 215   |       |
|                                                                                  | Edh at Noon | 32     | 41    | 45    | 51    | 51    | 48    | 47    | 41    | 35    | 33    | 29    | 27    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 500    | 797   | 1229  | 1810  | 2260  | 2365  | 2261  | 2082  | 1618  | 985   | 548   | 351   |       |
|                                                                                  | RadStd      | 80     | 77    | 85    | 89    | 72    | 111   | 126   | 69    | 63    | 77    | 56    | 56    |       |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page